

```

1 ;Created by Roger Williams 23/05/2021
2 ;
3 ;Saves portion of BASIC memory to disk
4 ;
5 ;REASON
6 ;
7 ;Allows saving of machine code AND basic to disk removing need to load
8 ;asm into memory via DATA statements!
9 ;
10 ;
11 ;PROCEDURE
12 ;
13 ;Create asm program IN the BASIC memory area (suggest 3048- as need mem to run
14 ;BASIC program to start the asm routines!)
15 ;
16 ;Calculate the last memory address the last asm routine is in, convert to
17 ;lo/hi bytes enter as the X/Y values in las asm statement in this program
18 ;
19 ;Generate a BASIC listing to load this asm routine into memory
20 ;NOTE: do not change THIS asm routines start address from 49152 as this is part
21 ;of your project!
22 ;
23 ;Copy the routine into VICE and run
24 ;MANUALLY delete all the BASIC lines
25 ;Generate BASIC lines for all your asm routines
26 ;Load each one into VICE and run loader then as above manually delete the
27 ;BASIC lines
28 ;Repeat until all asm routines are loaded into memory
29 ;Create a BASIC program to START your asm project e.g. SYS 3048
30 ;
31 ;NOTE the lowest mem start location for your asm routine(s) is the 2049 + TOTAL ;size of
the
32 ;LARGEST BASIC loader code
33 ;
34 ;e.g. say the asm routine was 1k then BASIC would need 2049 + the size of
35 ;BASIC DATA load routine, lets say it was 1.5K so the asm routines lowest
36 ;possible start address would be 2049 +1500 = 3549 say 3550 for safety
37 ;
38
39 *= $c000 ;49152
40
41 start
42
43     lda #4                ;length of file name
44     ldx #<tmp             ;low byte pointing to the file name
45     ldy #>tmp             ;high byte pointing to the file name
46     jsr $ffbd             ;SETNAM - c64 now knows the filename
47
48     lda #2                ;logical file number
49     ldx #8                ;floppy drive
50     ldy #1                ;secondary address, e.g. 15 to read the error channel
51     jsr $ffba             ;setlfs - open disk file for writing
52
53     lda #1                ;#low byte start address this figure is ALWAYS 1
54     sta $2b
55     lda #8                ;#high byte start address this figure is ALWAYS 8
56     sta $2c               ;represents start of BASIC mem - 2049
57     lda #$2b              ;zero-page index to the pointer
58     ldx #66               ;low byte (possibly +1) of END memory address
59     ldy #12               ;high byte of END memory address
60     jsr $ffd8             ;save ram to disk
61
62     lda #2                ;logical file number
63     jsr $ffc3             ;close file
64
65 exit
66     rts
67     brk
68

```

```
69 ;filename to save to NOTE: no filename extension required
70 tmp text "test",0
```